Brunda Marpadaga

Embedded Firmware Engineer | C/C++ | FreeRTOS | Embedded Linux | FPGA SoC Bring-up
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Education

Portland State University — Portland, OR
M.S. Electrical and Computer Engineering – ***GPA 3.85/4*** — *Graduated Dec 2025*

- **Coursework:** Embedded Systems Programming, Microprocessor System Design, Computer Architecture

Technical Skills

Languages: C, C++, SystemVerilog, MATLAB
Firmware & OS: FreeRTOS, bare-metal, embedded Linux (PetaLinux/Yocto, device tree)
Interfaces: I2C, SPI, UART, GPIO, AXI, Ethernet, TCP/IP, MQTT
Hardware: STM32, ESP8266/Arduino, Xilinx Zynq-7000 SoC (Zybo Z7-20), Artix-7 (Nexys A7-100T)
Debug & Tools: JTAG, J-Link/Segger Ozone, GDB, STM32CubeIDE, Vitis, CMake, Git, oscilloscope, logic analyzer

Projects

FPGA Accelerator + Dual-Core AMP Firmware on Zynq SoC | *C, ARM Asm, FreeRTOS, AXI DMA* code

- Delivered up to **9.4× speedup** over an ARM Cortex-A9 software baseline by offloading matrix multiply to a tiled FPGA accelerator (teammate wrote the HLS kernel; I built the firmware, data path, tests and instrumentation).
- Partitioned both Cortex-A9 cores for asymmetric multiprocessing (FreeRTOS + bare-metal) with a split memory map, separate linker script and shared-memory mailbox; wrote an LDREX/STREX + DMB inter-core spinlock.
- Found that coherent (ACP) DMA was 1.9–2.6× faster than non-coherent (HP0), and wrote a negative test proving a skipped cache flush silently corrupts data; used CP15 PMU and PL310 L2 counters to explain the speedup.

Embedded Linux Bring-up: I2C Drivers + MQTT/TLS Telemetry | *C, PetaLinux/Yocto, Device Tree* code

- Brought up embedded Linux on a Zybo Z7-20 from a Vivado hardware export: PetaLinux BSP, kernel config, device tree, U-Boot SD boot and serial console.
- Implemented register-level C interface libraries via `/dev/i2c` for BME280 and SSD1306 peripherals from datasheets, deploying a C MQTT/TLS client via BitBake recipe to publish telemetry to the cloud.
- Fixed a non-probing I2C kernel driver (unwired interrupt missing from the device tree), a 32-bit overflow caught with a host-side test, and TLS failures caused by the board's missing RTC.

RTOS Closed-Loop Control: Register-Level I2C Driver | *C, FreeRTOS, I2C, PWM, MicroBlaze*

- Wrote a register-level I2C driver for the TSL2561 light sensor from its datasheet (init, gain/timing, lux math).
- Structured the application as FreeRTOS tasks for sensing, PID control and PWM output to hold LED brightness at a setpoint; brought up the target as a MicroBlaze soft-core on a Nexys A7 (Artix-7 FPGA) and debugged over JTAG.

Interrupt-Driven IoT Infant Monitor | *C++, ESP8266, Wi-Fi/HTTP*

- Sampled 3 sensors on an ESP8266 using hardware interrupts, drove a servo actuator, and pushed telemetry over Wi-Fi to ThingSpeak with phone alerts via IFTTT webhooks.

Experience

Android Developer (Open Source Contributor) — Mar 2026 – Present
Greenstand — *Remote, U.S.*

- Built mobile features in Kotlin/Compose for the open-source Treetracker app, collaborating via Git reviews and GitHub Actions CI pipelines.

Associate Software Engineer — Jul 2022 – Dec 2023
Prodapt — *Hyderabad, India*

- Built full-stack web applications in Java Spring Boot and MongoDB in an Agile team using Git and Docker.

Failure Analysis Intern — May 2019 – Nov 2019
Electronics Corporation of India Limited (ECIL) — *Hyderabad, India*

- Debugged production Smart Energy Meter boards using test jigs, multimeter and JTAG to verify PCB hardware and firmware functionality, performed schematic-driven root-cause analysis, and fed corrective actions back to the soldering team.